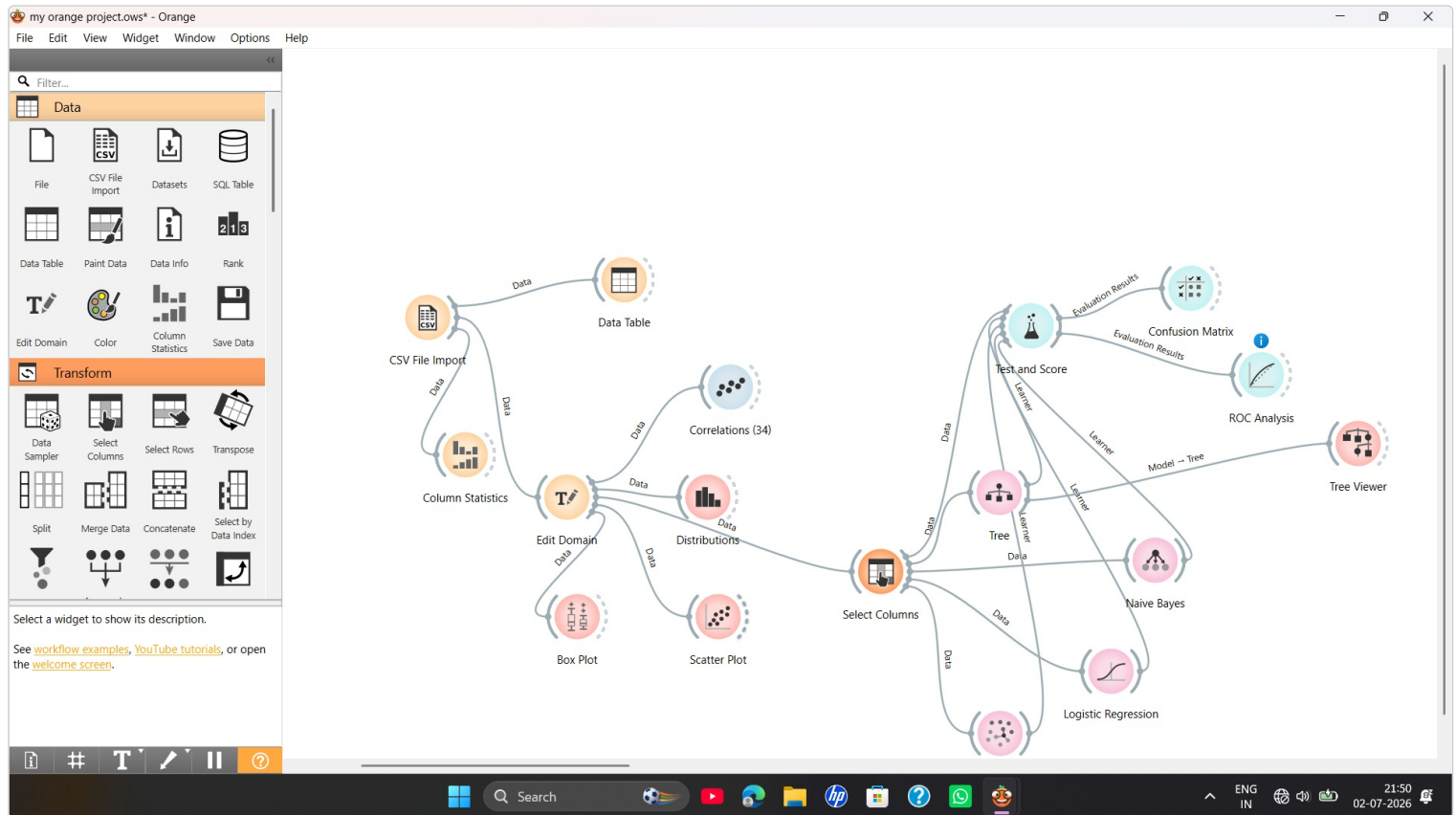




Screenshot Gallery — Heart Disease Prediction Project

12 screenshots | Orange Data Mining | July 2026

Full Canvas Overview



The complete Orange workflow: data loading, cleaning, exploratory widgets, and the four-model evaluation pipeline, all connected on one canvas.

Data Table

Data Table - Orange

File Edit View Window Help

Info

1025 instances (no missing data)
14 features
No target variable.
No meta attributes.

Variables

☒ Show variable labels (if present)
☐ Visualize numeric values
☒ Color by instance classes

Selection

Clear Selection

☒ Select full rows

Restore Original Order

☒ Send Automatically

	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	ca	thal
1	52	1		0	125	212	0	1	168	0	1	2	2
2	53	1		0	140	203	1	0	155	1	3.1	0	0
3	70	1		0	145	174	0	1	125	1	2.6	0	0
4	61	1		0	148	203	0	1	161	0	0	2	1
5	62	0		0	138	294	1	1	106	0	1.9	1	3
6	58	0		0	100	248	0	0	122	0	1	1	0
7	58	1		0	114	318	0	2	140	0	4.4	0	3
8	55	1		0	160	289	0	0	145	1	0.8	1	1
9	46	1		0	120	249	0	0	144	0	0.8	2	0
10	54	1		0	122	286	0	0	116	1	3.2	1	2
11	71	0		0	112	149	0	1	125	0	1.6	1	0
12	43	0		0	132	341	1	0	136	1	3	1	0
13	34	0	1		118	210	0	1	192	0	0.7	2	0
14	51	1		0	140	298	0	1	122	1	4.2	1	3
15	52	1		0	128	204	1	1	156	1	1	1	0
16	34	0	1		118	210	0	1	192	0	0.7	2	0
17	51	0	2		140	308	0	0	142	0	1.5	2	1
18	54	1		0	124	266	0	0	109	1	2.2	1	1
19	50	0	1		120	244	0	1	162	0	1.1	2	0
20	58	1	2		140	211	1	0	165	0	0	2	0
21	60	1	2		140	185	0	0	155	0	3	1	0
22	67	0		0	106	223	0	1	142	0	0.3	2	2
23	45	1		0	104	208	0	0	148	1	3	1	0
24	63	0	2		135	252	0	0	172	0	0	2	0
25	42	0	2		120	209	0	1	173	0	0	1	0
26	61	0		0	145	307	0	0	146	1	1	1	0
27	44	1	2		130	233	0	1	179	1	0.4	2	0
28	58	0	1		136	319	1	0	152	0	0	2	2
29	56	1	2		130	256	1	0	142	1	0.6	1	1

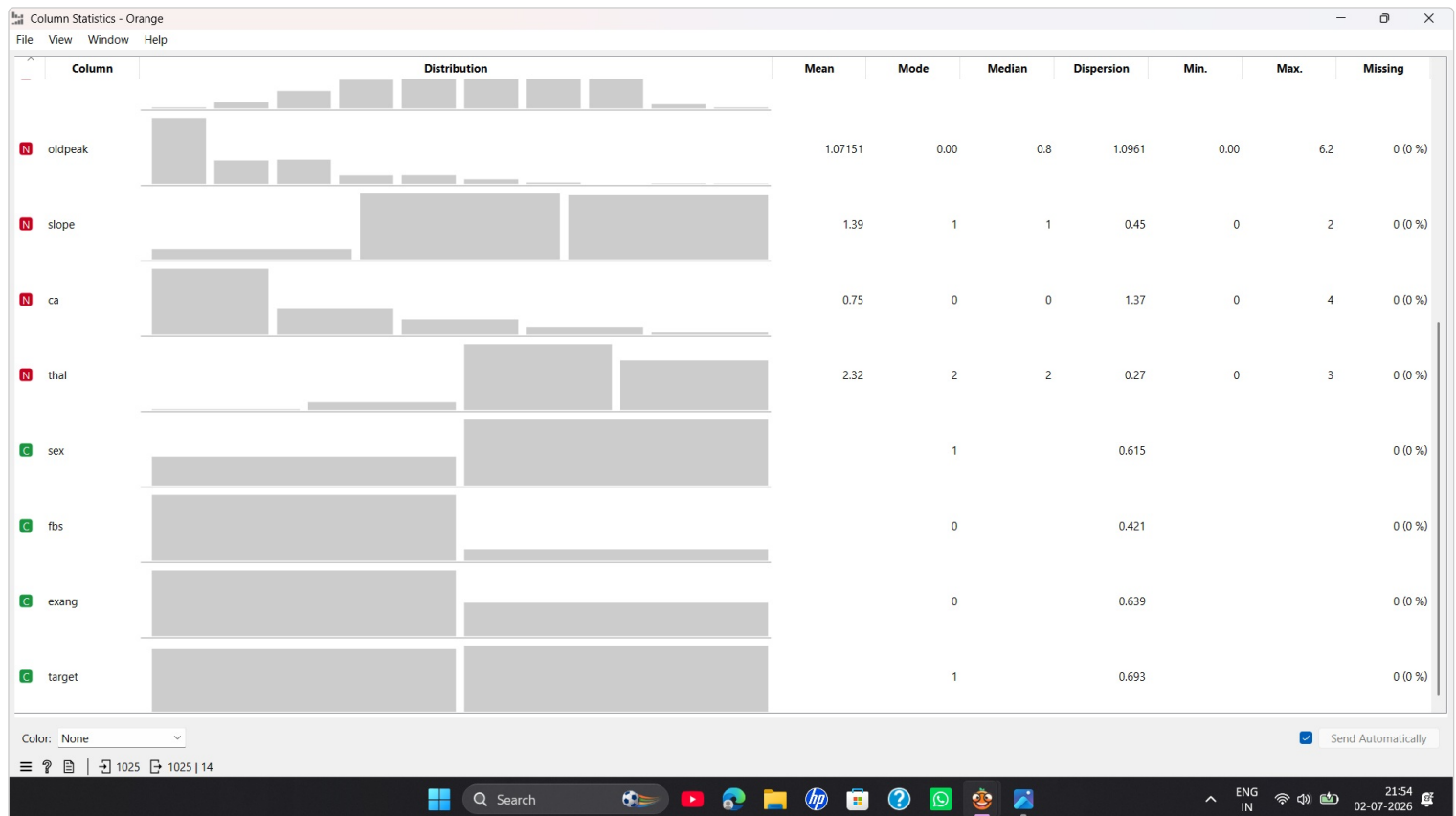
Raw view of the 1,025-row heart.csv dataset after import, showing all 14 columns.

Edit Domain

The screenshot shows the Orange3 software interface. The main window is titled "my orange project.ows* - Orange". The "Edit Domain" dialog box is open, displaying a list of variables on the left and an "Edit" panel on the right. The variable "age" is selected in the list, and its type is set to "Numeric". The "Edit" panel also shows a "Labels" section with "Key" and "Value" columns. The background workflow includes several widgets: "Data Table", "Edit Domain", "Data Sampler", "Select Columns", "Select Rows", "Transpose", "Split", "Merge Data", "Concatenate", "Select by Data Index", "Learner", "Model -> Tree", "Naive Bayes", "Logistic Regression", "kNN", "Evaluation Results", "Confusion Matrix", "ROC Analysis", and "Tree Viewer".

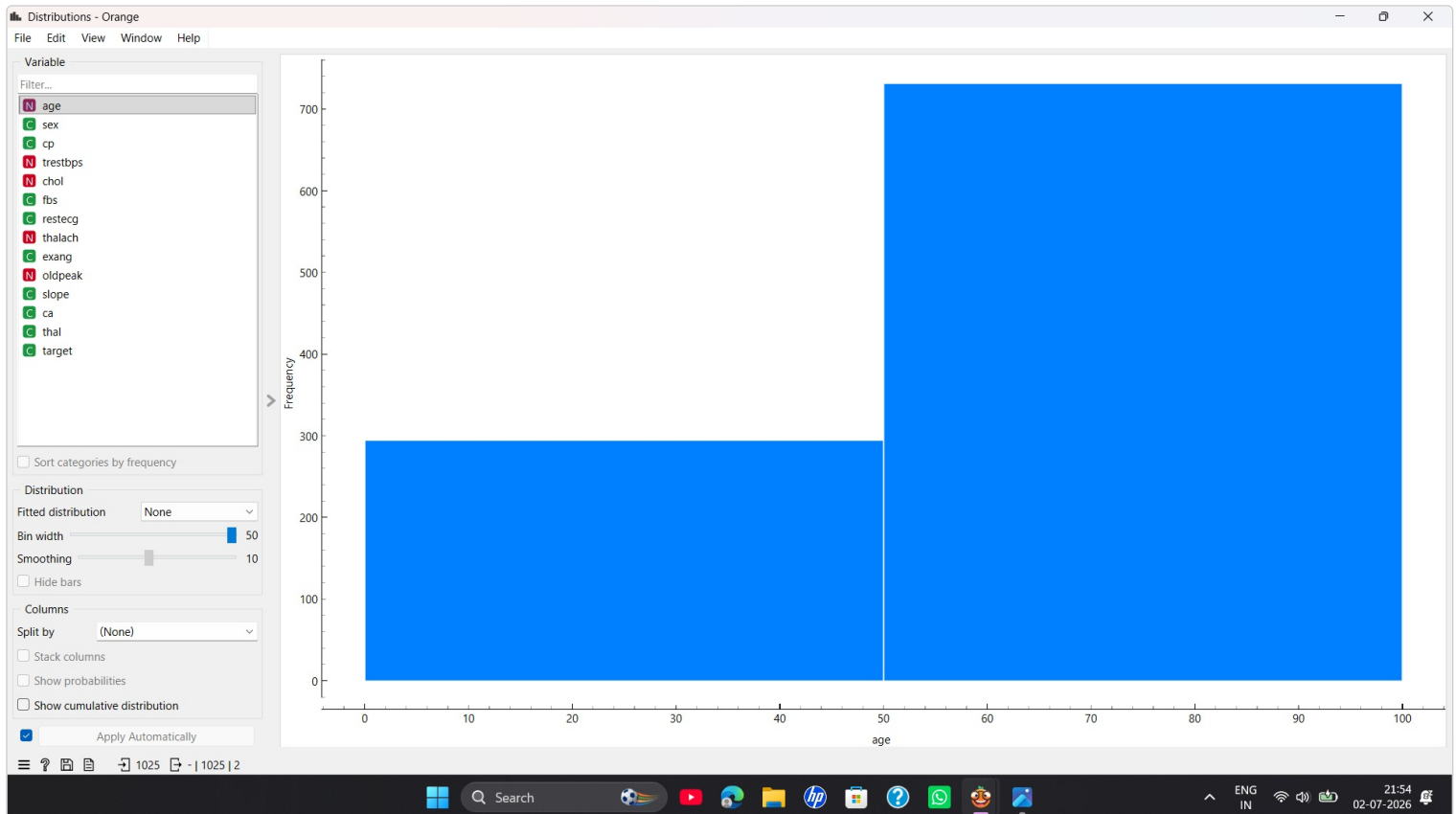
Re-typing coded columns (cp, restecg, slope, ca, thal) from numeric to categorical.

Column Statistics



Mean, mode, median, dispersion, and range for every feature in the dataset.

Distributions — Age



Histogram of patient age across the full dataset.

Box Plot — Age



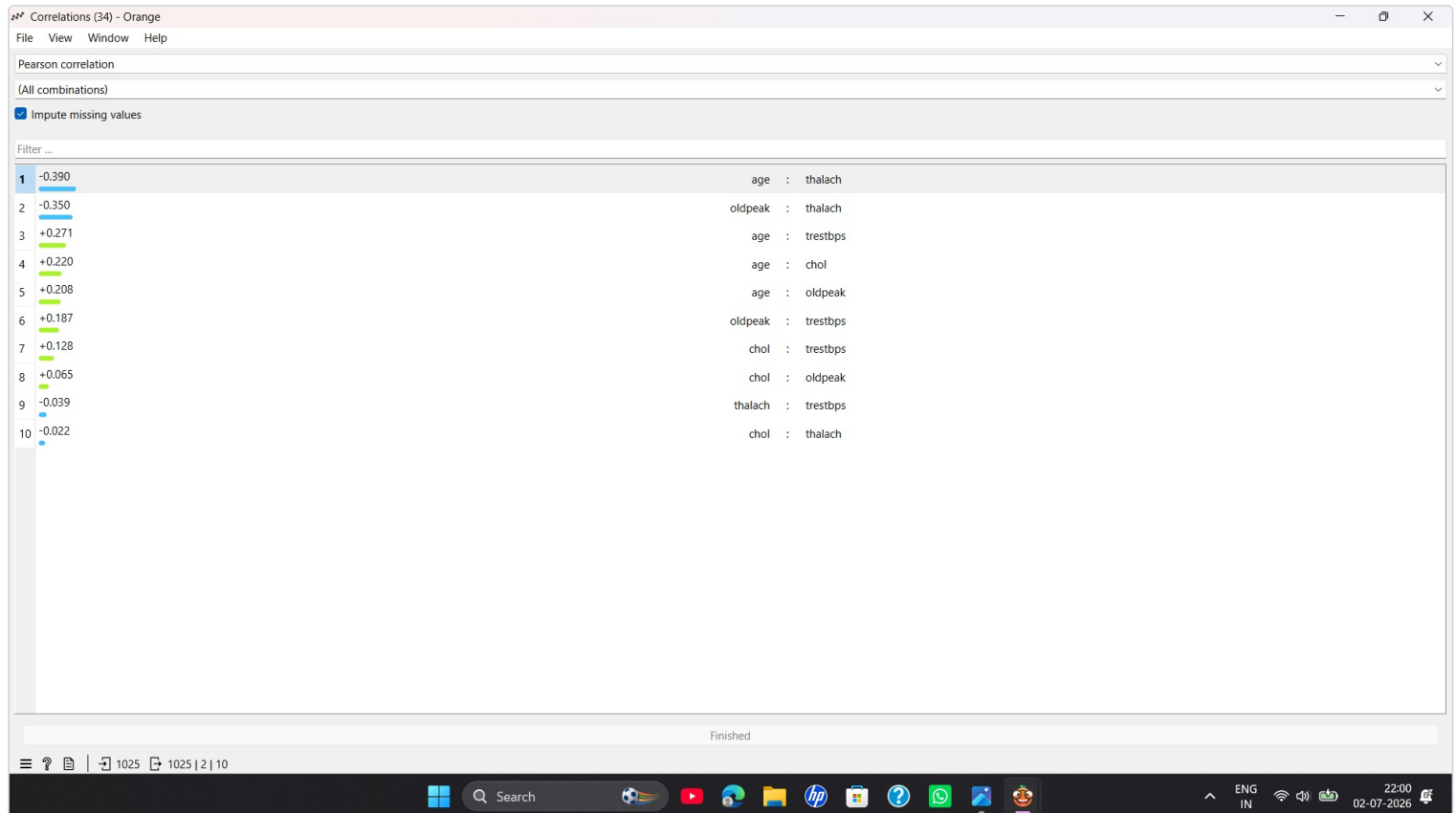
Age spread: mean 54.4 ± 9.1 years, median 56, IQR 48-61.

Scatter Plot



Age vs. sex, shaped by exercise-induced angina and sized by resting blood pressure.

Correlations



Pearson correlation ranking across all numeric feature pairs.

Test and Score

Test and Score - Orange

File Edit View Window Help

Cross validation

Number of folds: 5

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Tree	0.986	0.957	0.957	0.957	0.957	0.914
Naive Bayes	0.919	0.846	0.846	0.846	0.846	0.691
Logistic Regression	0.937	0.866	0.866	0.867	0.866	0.733
kNN	0.864	0.726	0.726	0.727	0.726	0.452

Compare models by: Area under ROC curve

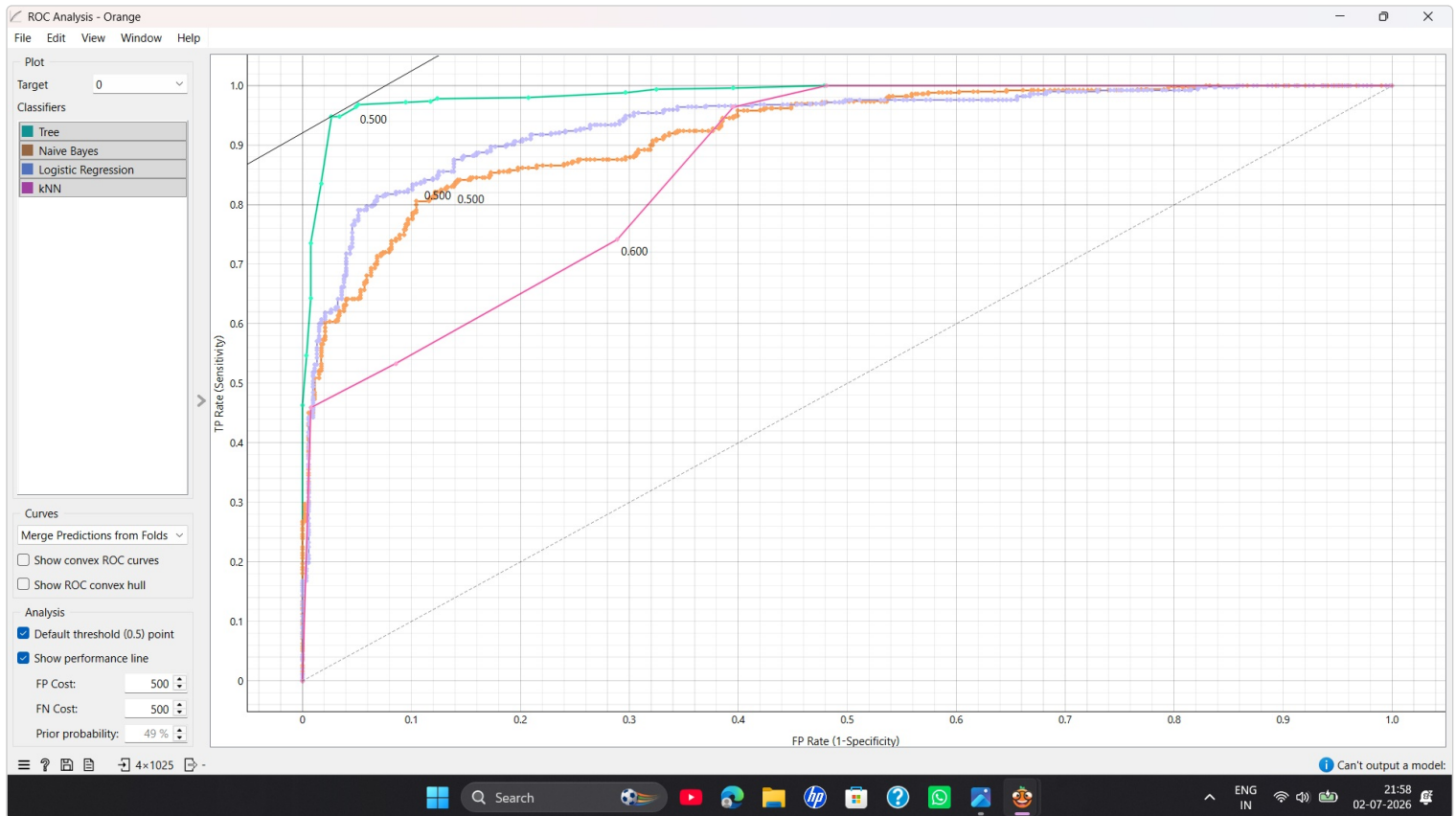
☐ Negligible diff.: 0.1

Tree	Tree	Naive Bayes	Logistic Regression	kNN
Naive Bayes	0.000	1.000	0.996	1.000
Logistic Regression	0.004	0.944	0.056	0.995
kNN	0.000	0.005	0.005	0.995

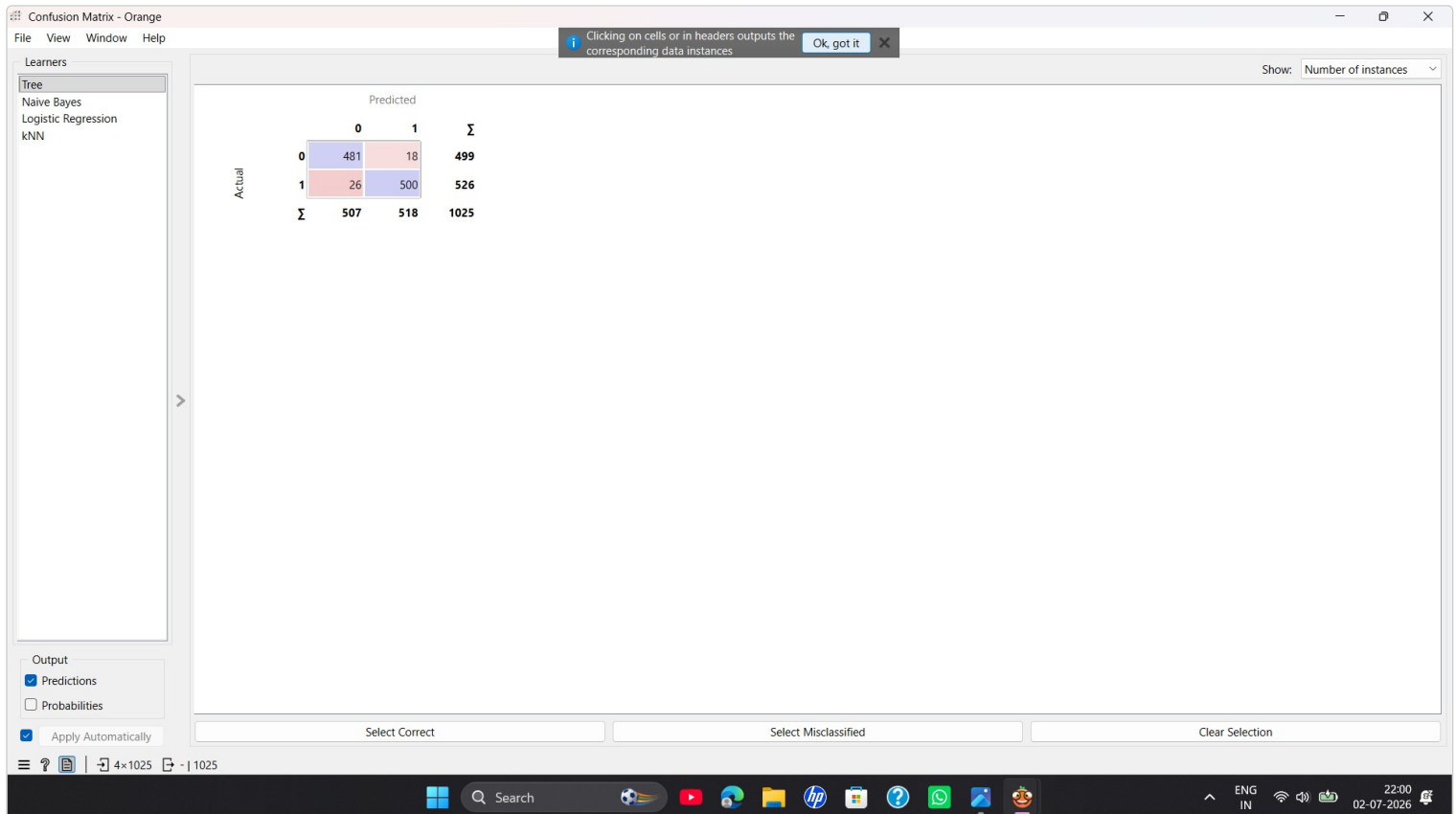
Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Cross-validated comparison of Tree, Naive Bayes, Logistic Regression, and kNN — Tree wins with AUC 0.986.

ROC Analysis



Confusion Matrix — Tree



481 true negatives, 500 true positives, only 44 total misclassifications out of 1,025.



The decision tree's actual splits — cp, thal, and ca dominate the top of the tree.